DOĞRULAR VE AÇILAR

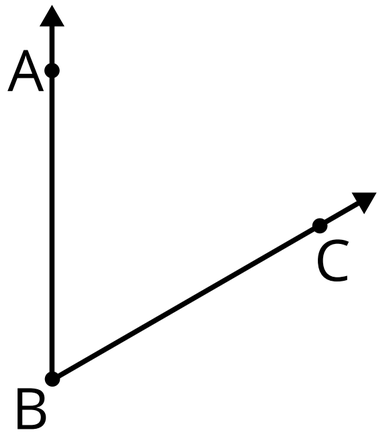
ProblemDede

M.7.3.1.1. Bir açıyı iki eş açıya ayırarak açıortayı belirler.

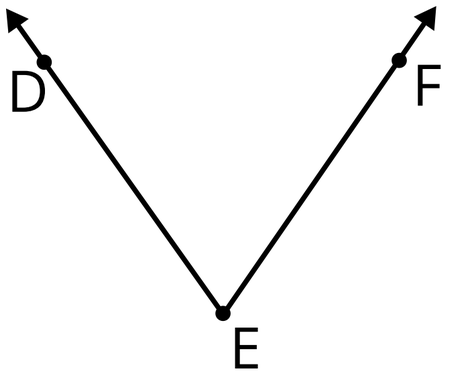
Dinamik geometri yazılımlarından yararlanılabilir.

ProblemDede

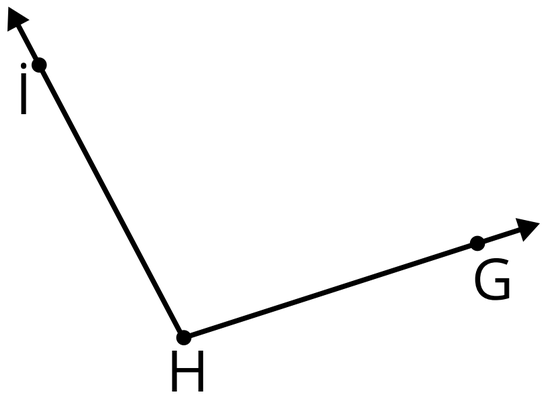
1. Aşağıda verilen açıları iki eşit parçaya ayıracak şekilde ışınlar çiziniz. Oluşan yeni açıların ölçülerini sembolle gösteriniz.



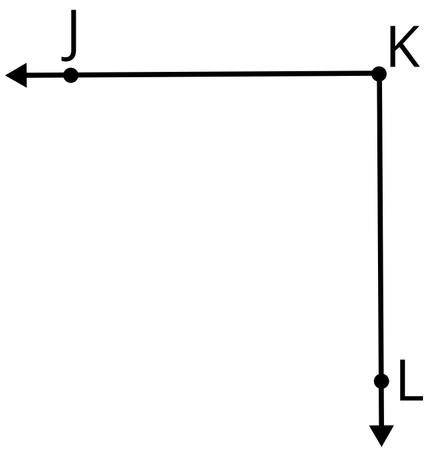
$s(\widehat{ABC}) = 60°$



$s(\widehat{DEF}) = 70°$



$s(\widehat{GH\dot{I}}) = 100°$

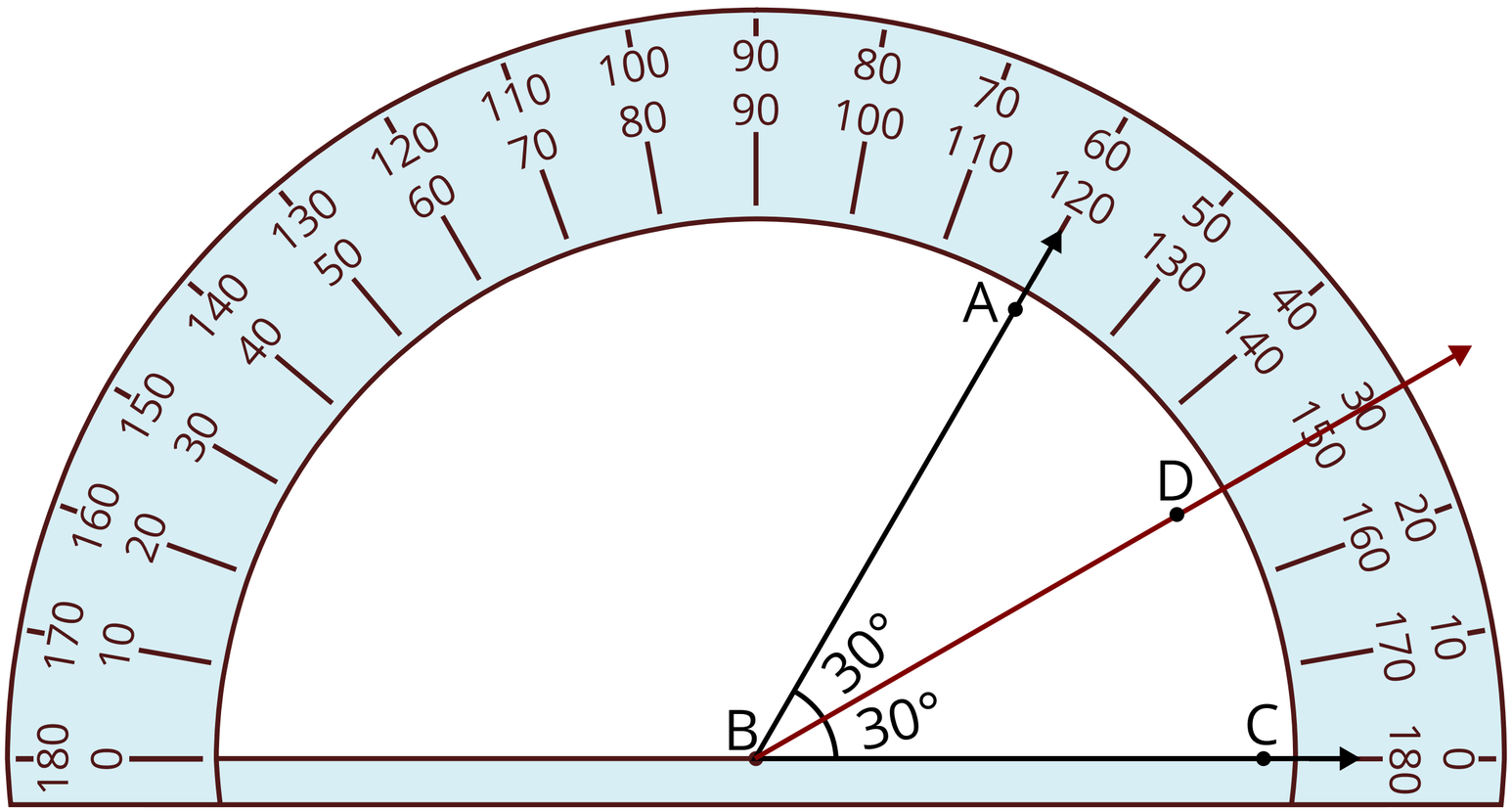


$s(\widehat{JKL}) = 90°$

ProblemDede

BİLGİ

Bir açının ölçüsünü iki eşit parçaya bölen ışına, o açının **açıortayı** denir.



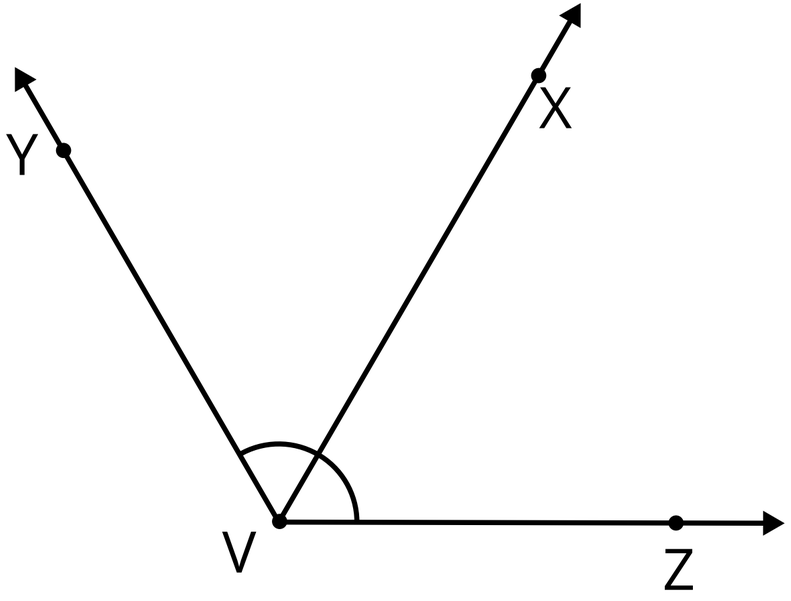
[BD, 60°'lik $\widehat{ABC}$ 'sını iki eş parçaya ayırmıştır.

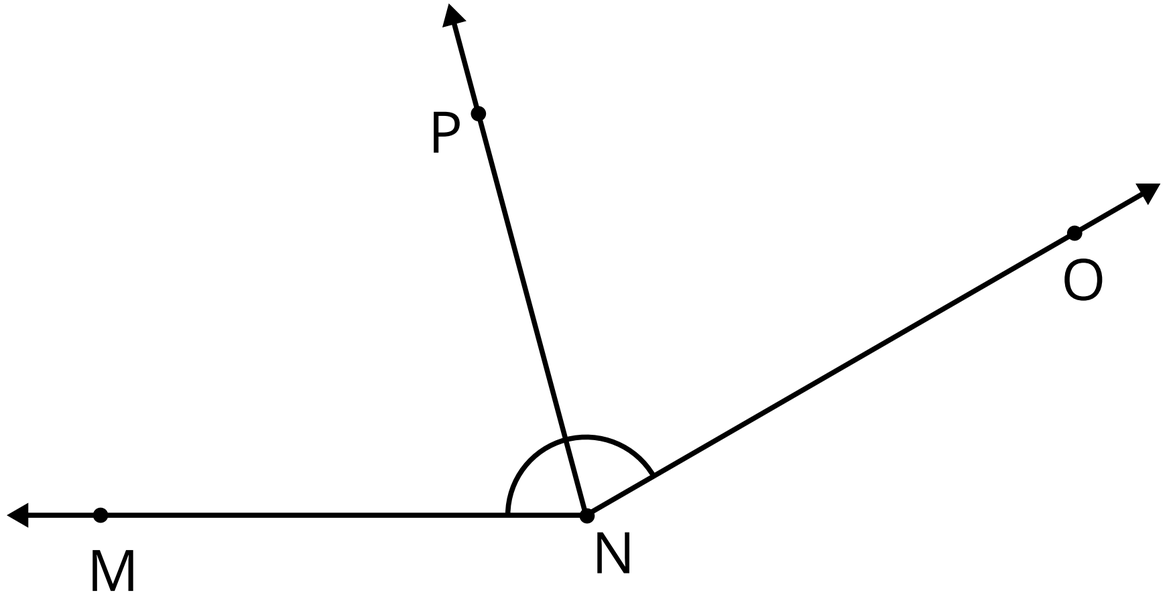
$s(\widehat{ABD}) = s(\widehat{DBC}) = 30°$

[BD, $\widehat{ABC}$'sının açıortayıdır.

ProblemDede

2. Aşağıdaki verilere göre istenen açıları bulunuz.





a. [VX açıortay

$s(\widehat{YVX}) = 60°$

$s(\widehat{XVZ})$ = ...

$s(\widehat{YVZ})$ = ...

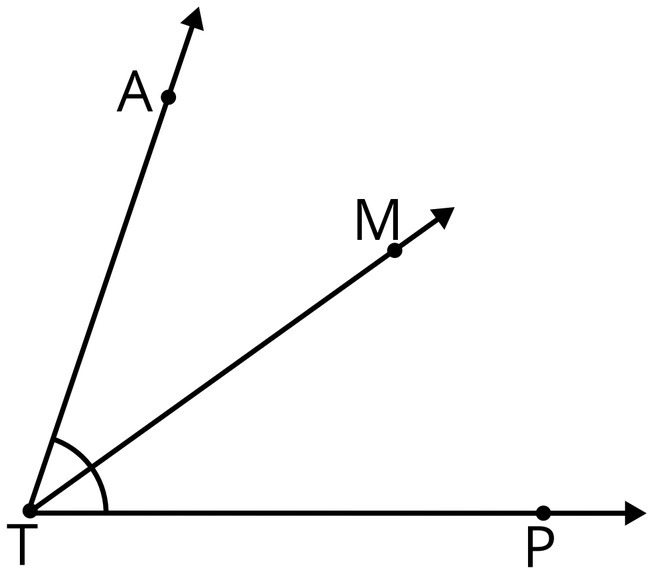
b. [NP açıortay

$s(\widehat{MNO})$ = ...

$s(\widehat{MNP}) = 75°$

$s(\widehat{PNO})$ = ...

ProblemDede

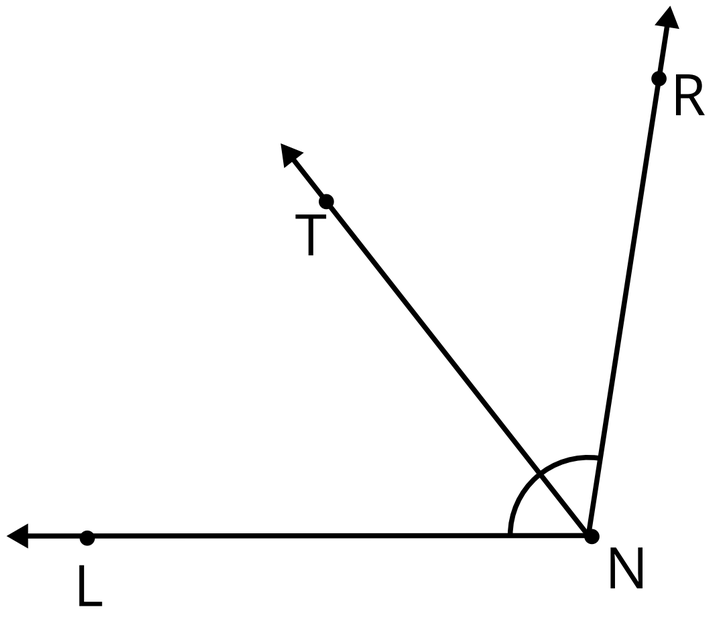


c. [TM açıortay

$s(\widehat{ATM}) = 32{,}5°$

$s(\widehat{MTP})$ = ...

$s(\widehat{ATP})$ = ...



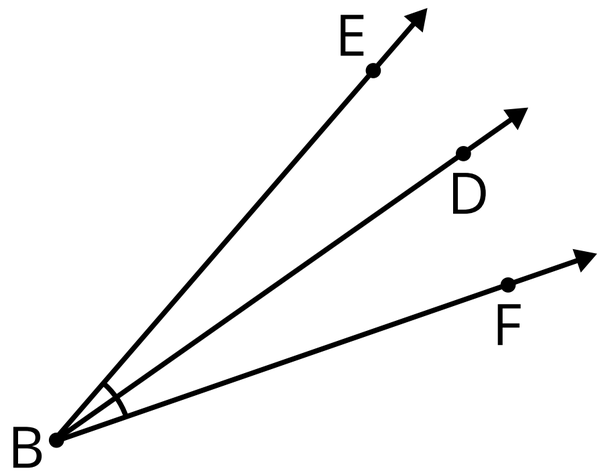
ç. [NT açıortay

$s(\widehat{LNR}) = 99°$

$s(\widehat{LNT})$ = ...

$s(\widehat{TNR})$ = ...

ProblemDede

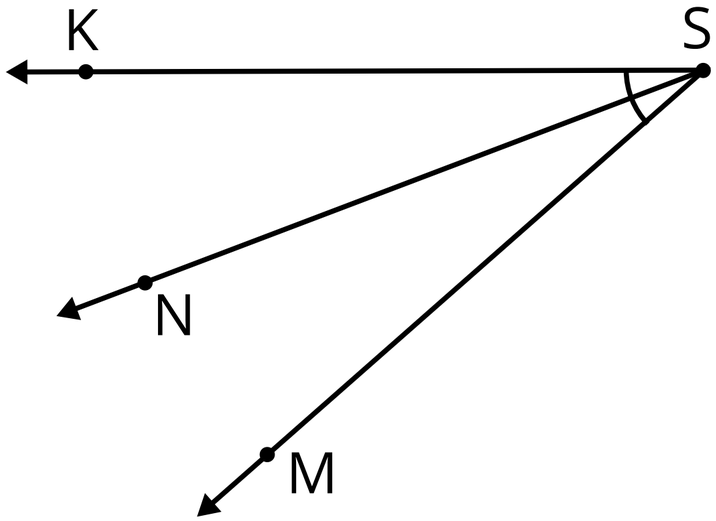


d. [BD açıortay

$s(\widehat{EBF}) = 28{,}5°$

$s(\widehat{EBD})$ = ...

$s(\widehat{DBF})$ = ...

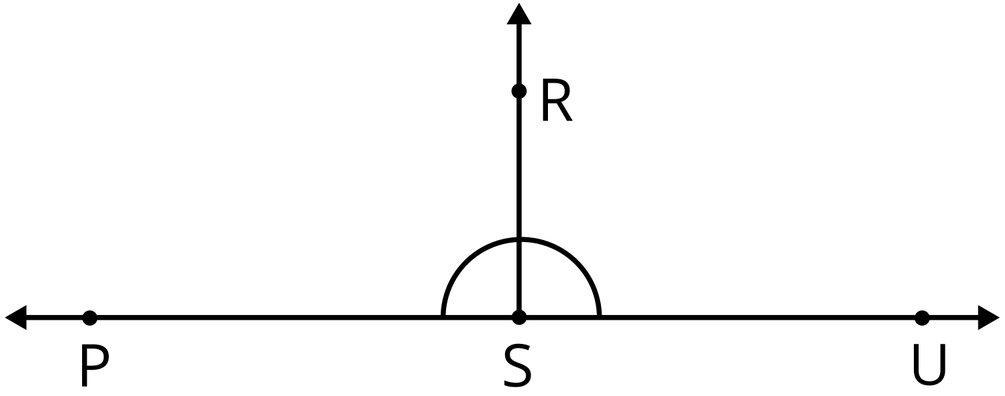


f. [SN açıortay

$s(\widehat{NSM}) = 20{,}7°$

$s(\widehat{KSM})$ = ...

$s(\widehat{KSN})$ = ...



e. [SR açıortay

$s(\widehat{PSR}) = 90°$

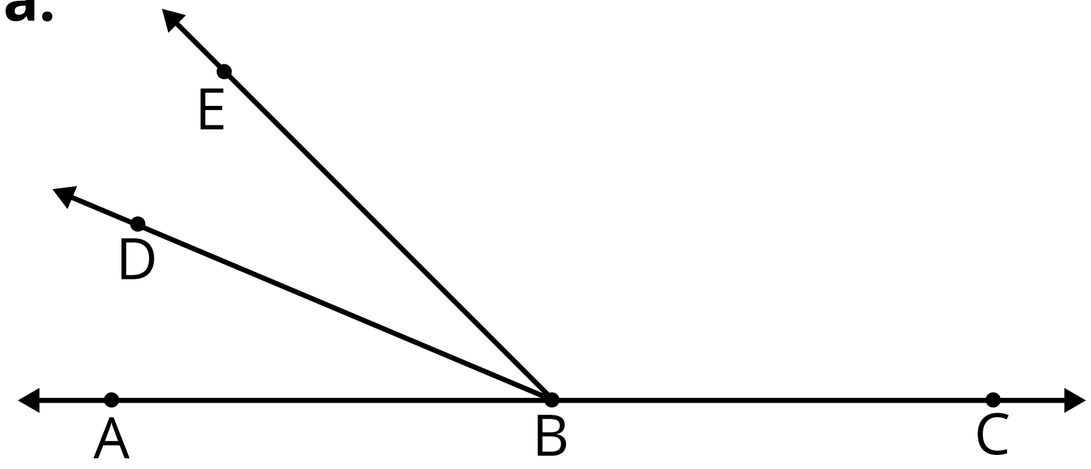
$s(\widehat{RSU})$ = ...

$s(\widehat{PSU})$ = ...

ProblemDede

3. Aşağıda verilen soruları çözünüz.

a.



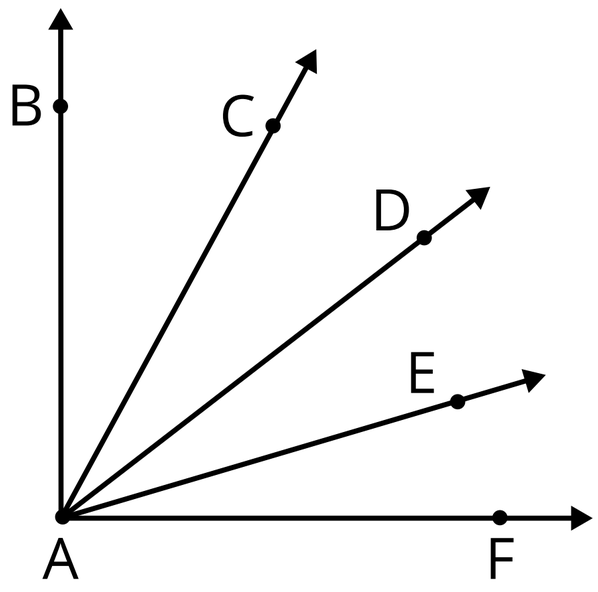
$\widehat{ABC}$ doğru açıdır. [BD ışın $\widehat{ABE}$ sının açıortayıdır. s($\widehat{ABD}$) = 32° ise

s($\widehat{DBE}$) = ...

S($\widehat{EBC}$) = ...

ProblemDede

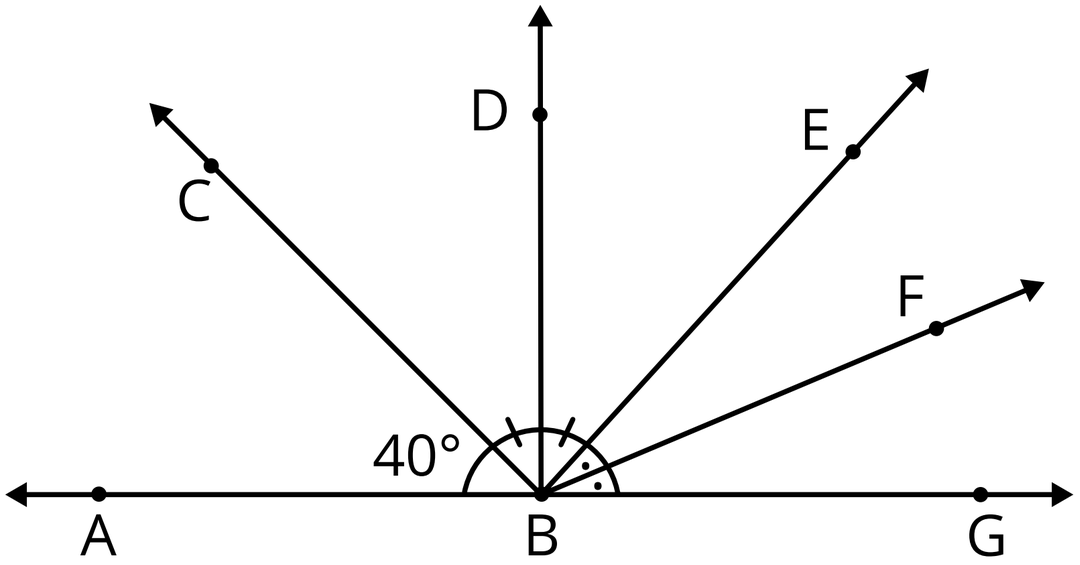
b.



[AC, $\widehat{BAD}$ nın açıortayı; [AE, $\widehat{DAF}$ nın açıortayıdır. s($\widehat{BAC}$) = 30°, s($\widehat{EAF}$) = 15° ise $\widehat{BAF}$ açısının açıortayı ile [AD ışını arasındaki açı kaç derecedir?

ProblemDede

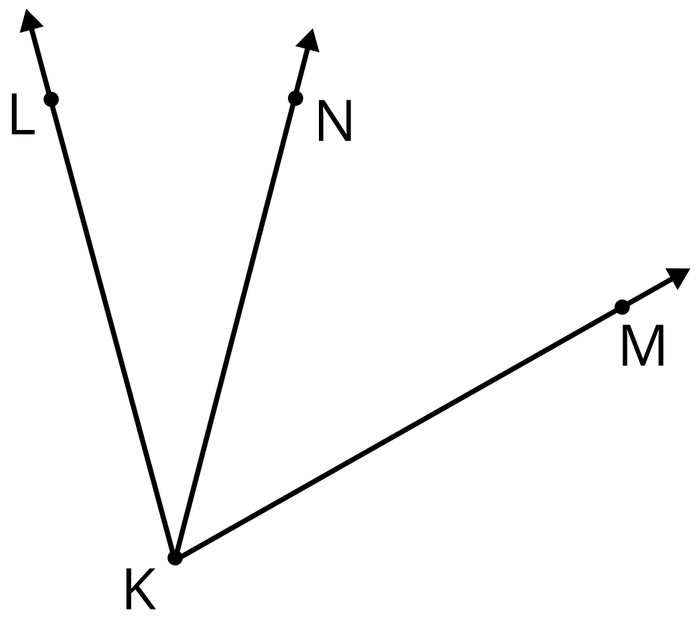
c.



Şekle göre AG bir doğru ise $s(\widehat{DBE}) + s(\widehat{EBF})$ kaç derecedir?

ProblemDede

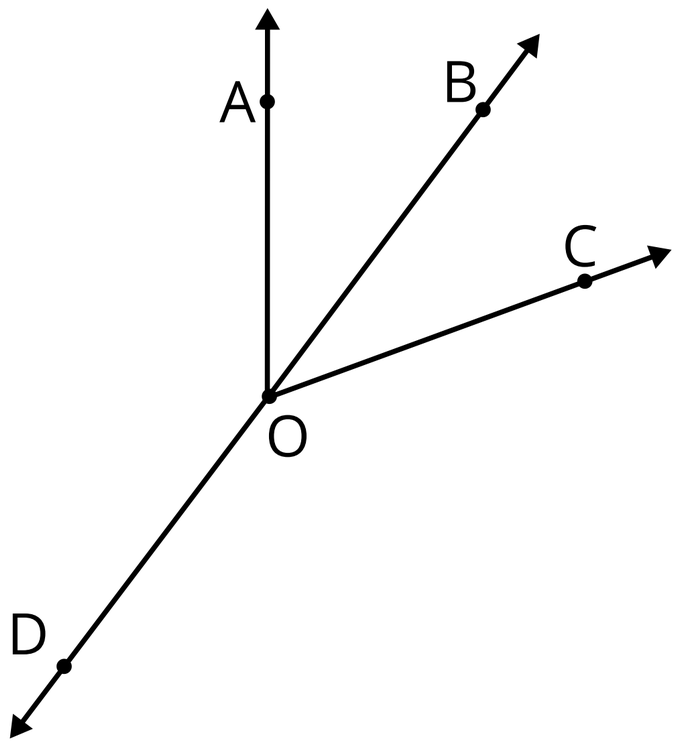
ç.



$s(\widehat{LKM}) = 48°$ ve $s(\widehat{MKN}) = 29°$ dir. $\widehat{LKM}$ ve $\widehat{MKN}$ açılarının açıortayları arasında kalan açı kaç derecedir?

ProblemDede

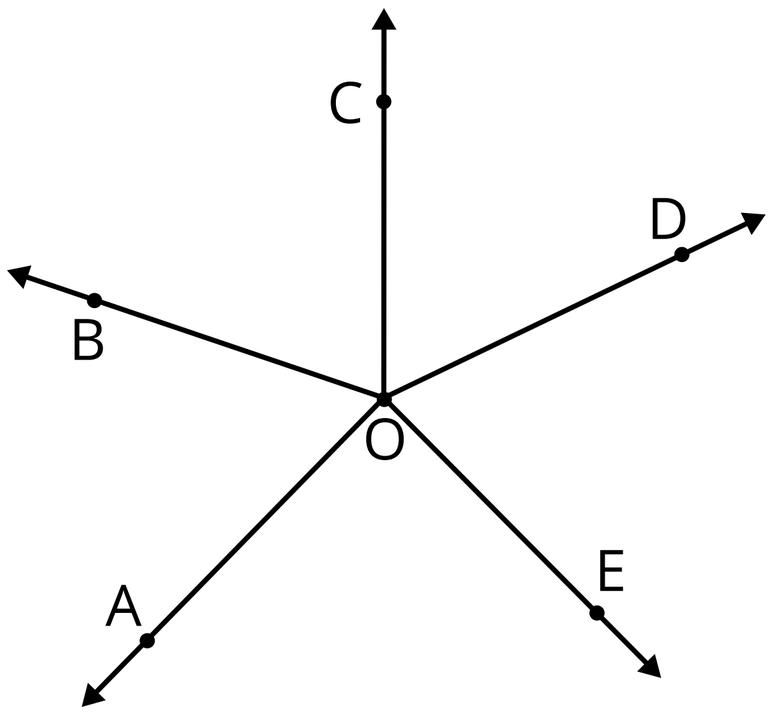
d.



DB doğrusu $\widehat{AOC}$ açısının açıortayıdır. $S(\widehat{AOB}) = 33{,}5°$ ise $\widehat{DOC}$ açısının ölçüsü kaç derecedir?

ProblemDede

e.



[OB ve [OD sırası ile $\widehat{AOC}$ ve $\widehat{COE}$ açılarının açıortaylarıdır. s($\widehat{COD}$) = 72,25° ve s($\widehat{AOB}$) = 35° ise s($\widehat{AOE}$) kaç derecedir?

ProblemDede